

A Second-Order Conditional Singular Series for Twin-Prime Gaps on the $6N$ Skeleton: the Spatial-Compression Penalty and a Residual Shield-Gain Term

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Abstract

Part III gave a first-order conditional singular series $\mathfrak{S}_{2,\omega}(d) \approx C_2(d)L_\omega(d)$ for the twin-prime gap distribution on the $6N \pm 1$ skeleton, reproducing the ω -trends of the gaps 30, 42, 60 and the direction of the collapse of 210, but leaving a stable high- ω residual on 210 (observed 0.41 where the first-order series predicts 0.62 at $\omega = 6$). Here we identify the second-order term responsible. The first-order series treats the primes $q \nmid N$ by their unconditional Hardy–Littlewood factor; but conditioning on $q \nmid N$ is itself a constraint—it removes the residue $N \equiv 0$ and crowds N into the remaining $q-1$ classes, where the same fatal residues now occupy a larger fraction. This *spatial-compression penalty* $\text{pen}_q(d) = \frac{(q-1-k_q)/(q-1)}{(q-k_q)/q} < 1$ lowers the survival of factor-rich centres on gaps locked by small primes. Evaluated on the 23,988,173 twin centres of S_{10} with each centre’s real factor set, the second-order series closes the 210 residual essentially completely (at $\omega = 6$, observed 0.413, second order 0.426, against first order 0.357), across all strata. It also corrects the direction on 42 but only partially: about half of the high- ω rise of 42 remains unexplained, an effect we attribute to a complementary *factor-coincidence shield gain* on the $q \mid N$ side (the gap 42 contains the factor 7) that the compression penalty does not capture. The long-gap residual of Part III is thus resolved; the short-gap residual is sharpened into a named two-centre open problem. No claim is made about the infinitude of twin primes or any k -tuple conjecture.

1 The first-order blind spot

Part III [3] wrote the first-order conditional singular series as $\mathfrak{S}_{2,\omega}(d) \approx C_2(d)L_\omega(d)$, where for each prime $q \mid N$ the congruence-lockdown rule forbids the gap residues $d \equiv \pm 6^{-1} \pmod{q}$, and each prime $q \nmid N$ contributes its *unconditional* Hardy–Littlewood factor. The residual on 210 showed this last step to be incomplete: at high ω the observed preference fell well below the first-order prediction.

The blind spot is the treatment of $q \nmid N$. In a stratum of factor-rich centres, “ $q \nmid N$ ” is not a neutral background event but a constraint. Fix a prime q that does not lock the gap d (so $q \nmid 6d \pm 1$, and the fatal residue set of d modulo q does not contain 0). Unconditionally, N ranges over all q classes and the joint twin survival omits the k_q fatal residues, giving a factor $\propto (q - k_q)/q$. But conditioning on $q \nmid N$ deletes the class $N \equiv 0$: N now ranges over only $q-1$ classes, while the same k_q fatal residues persist (none of them is 0). The conditional survival is therefore $\propto (q - 1 - k_q)/(q - 1)$.

Proposition 1 (spatial-compression penalty). *For a prime q that does not lock the gap d , conditioning on $q \nmid N$ multiplies the per- q twin-survival factor by*

$$\text{pen}_q(d) = \frac{(q-1-k_q)/(q-1)}{(q-k_q)/q} = \frac{q(q-1-k_q)}{(q-1)(q-k_q)} < 1,$$

where k_q is the number of fatal residues of d modulo q (the classes $N \equiv \pm 6^{-1}, -d \pm 6^{-1}$, none equal to 0 since q does not lock d).

The penalty is strictly below 1: forced off the class $N \equiv 0$, the centre is crowded into a smaller space where the fatal residues are proportionally denser, so its twin survival drops. This is the second-order effect omitted at first order.

2 Second-order series and confrontation with S_{10}

Including the penalty, the second-order conditional weight of a gap d at a real centre N with small-prime factor set Q is

$$w^{(2)}(d, N) = C_2(d) \cdot \mathbf{1}\{\text{no } q \in Q \text{ locks } d\} \cdot \prod_{q \notin Q, q \nmid (6d \pm 1)} \text{pen}_q(d).$$

We evaluate this over the 23,988,173 twin centres of S_{10} using each centre's true factor set (no independence or uniform-combination assumption), aggregate by $\omega_{>3}(N)$, and compare to the observed preference and to the first-order series (Fig. 1, Table 1).

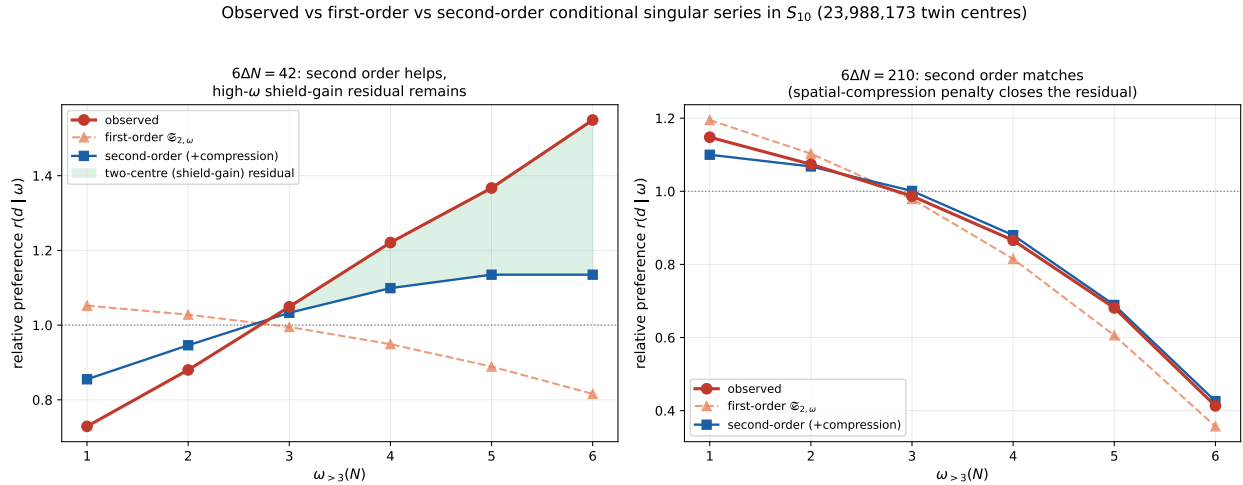


Figure 1: Observed (red), first-order (orange dashed), and second-order (blue) conditional singular series in S_{10} . **Right ($6\Delta N = 210$):** the spatial-compression penalty closes the first-order residual at every stratum. **Left ($6\Delta N = 42$):** the penalty corrects the direction and recovers about half of the high- ω rise, but a residual (shaded) remains, attributed to a complementary shield-gain term.

Long gap 210: resolved. The lock primes of 210 are 11, 19 (since $6 \cdot 35 - 1 = 209 = 11 \times 19$). At high ω , surviving centres are precisely those forced to avoid 11 and 19, so the compression penalty acts on exactly these primes ($\text{pen}_{11} \approx 0.94$). The second-order series reproduces the observed collapse at every stratum (Table 1, top), closing the Part III residual.

$6\Delta N$	model	$\omega=1$	2	3	4	5	6
210	observed	1.15	1.07	0.99	0.87	0.68	0.41
	first-order	1.20	1.10	0.98	0.82	0.61	0.36
	second-order	1.10	1.07	1.00	0.88	0.69	0.43
42	observed	0.73	0.88	1.05	1.22	1.37	1.55
	first-order	1.05	1.03	1.00	0.95	0.89	0.82
	second-order	0.86	0.95	1.03	1.10	1.14	1.14

Table 1: Observed, first-order, and second-order $r(d \mid \omega)$ in S_{10} . For 210 the second order tracks the observation at every stratum; for 42 it corrects the direction but stalls near 1.14 while the observation continues to 1.55.

Short gap 42: corrected in direction, residual remains. The lock primes of 42 are 41, 43 (both prime, since $6 \cdot 7 \mp 1 = 41, 43$), which are rarely divisors of a centre, so 42 suffers little lockdown and little compression—correctly, the second-order series turns the (wrongly decreasing) first-order curve upward. But it stalls near 1.14 while the observation reaches 1.55 (Table 1, bottom). About half of the high- ω rise is still unexplained.

Normalisation note. The relative preferences $r(d \mid \omega)$ here are normalised *model-internally*: each of the three curves (observed, first-order, second-order) is divided by its own ω -merged mean, so that the average over strata is unity. This differs from the global $C_2 \times$ envelope baseline used in Part III; the two conventions differ by a constant scaling per gap and per model, which shifts the absolute level but not the gradient in ω or the relative size of the residuals. We use the model-internal convention because the object of interest here is precisely the per-stratum departure of each model from the observation, which it isolates cleanly; readers comparing absolute $\omega = 1$ levels across Parts III and IV should account for this constant offset.

3 The residual: the single-centre limitation

It is important to be exact about what the second-order series is and is not. Proposition 1 is a *single-centre* condition: it conditions on the factorisation of the left centre N , and treats the right centre $N + d$ as independent of N given that conditioning. This is an idealisation. The two centres differ by a fixed d , so modulo any prime q their residues are *strongly correlated* ($N + d \equiv N + d$, deterministically), not independent. The genuinely missing piece is therefore not “a further order” on top of an exact formula, but the *two-centre correlation* that the single-centre second-order series omits by construction.

This omission is mild for 210 and severe for 42, which is exactly the pattern observed. For 210, the lock primes 11, 19 rarely divide both N and $N + 210$, so the single-centre treatment is close to correct and the residual is closed (Table 1). For 42 the correlation is strong and favourable: $42 = 6 \cdot 7$, and 7 is one of the small primes a factor-rich centre typically carries. When $7 \mid N$, both centres are shielded against 7 simultaneously, since $6N \pm 1 \equiv \pm 1$ and $6(N + 42) \pm 1 \equiv \pm 1 \pmod{7}$ alike—a *factor-coincidence shield gain* in which the gap’s own factor 7 aligns with the centre’s. The single-centre series, conditioning only on N , does not carry this joint two-centre shielding, and so under-predicts the rise of 42.

Open problem (two-centre correlation). The present second-order series is limited to a single-centre condition. Derive the two-centre conditional singular series that retains

the correlation between the factorisations of N and $N + d$ modulo each q — in particular the factor-coincidence shield gain when a prime dividing the centre also divides the gap (e.g. $7 \mid N$ for $d = 7$) — and test whether it accounts for the residual rise of 42 (observed 1.55 versus single-centre second-order 1.14 at $\omega = 6$ in S_{10}). The shield-gain mechanism above is at present only a *qualitative* hypothesis; unlike the compression penalty it has not been reduced to a verified closed-form factor. Establishing it quantitatively would complete the conditional singular series beyond the single-centre approximation.

4 Conclusion

We identified the second-order term of the conditional twin-gap singular series: a spatial-compression penalty arising because conditioning on $q \nmid N$ crowds the centre into $q - 1$ classes. Evaluated on 2.4×10^7 twin centres, it closes the long-gap (210) residual that Part III left open, at every stratum. It corrects the direction of the short-gap (42) trend but recovers only about half of its high- ω rise. The remainder is not a missing “further order” on top of an exact formula: the second-order series is by construction a *single-centre* condition, conditioning on the factorisation of N while treating $N + d$ as independent. The two centres are in fact correlated modulo each prime, and the unexplained rise of 42 reflects a favourable two-centre coincidence (the gap’s factor 7 shielding both centres when $7 \mid N$), which the single-centre series omits. We pose this two-centre conditional singular series, and the quantitative form of the shield gain, as the open problem. The picture is layered and explicit: first-order lockdown (absolute exclusion when $q \mid N$ locks the gap), second-order single-centre compression (survival drop when $q \nmid N$), and a remaining two-centre correlation (the shield gain, still only a qualitative hypothesis). As throughout this series, we make no claim about the infinitude of twin primes or any k -tuple conjecture; this is a conditional, factor-resolved refinement of the Hardy–Littlewood gap heuristic, demonstrated empirically and offered with its open remainder.

Data and code availability. The scan, the first- and second-order series evaluation, and the figure code are openly available [5]; the S_{10} twin count 23,988,173 matches Part I.

References

- [1] R. Chen, *Prime distribution on the $6N$ skeleton* (Part I), Zenodo, doi:10.5281/zenodo.20470367 (2026).
- [2] R. Chen, *Prime gaps on the $6N$ skeleton: an ω -dependent shift and a congruence-lockdown mechanism* (Part II), Zenodo, doi:10.5281/zenodo.20477664 (2026).
- [3] R. Chen, *A first-order conditional singular series for twin-prime gaps on the $6N$ skeleton, and its second-order residual* (Part III), Zenodo, doi:10.5281/zenodo.20498668 (2026).
- [4] G. H. Hardy, J. E. Littlewood, *Partitio Numerorum III*, Acta Math. **44** (1923), 1–70.
- [5] R. Chen, *$6N$ twin-prime conditional singular series: data and code*, <https://github.com/Ruqing1963/6N-twin-prime-second-order-residual> (2026).